

Feedback stabilisation for the McKean-Vlasov equation

Controlling the long-time behavior of interacting particle systems

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Ceilidh Dance

From local rules to global behaviour

- A traditional Scottish dance with a caller who steers the choreography (pattern).
- Local interactions lead to global co-ordination
- We model the agents with stochastic differential equations (SDEs)
- In the limit of many agents, the probability density follows a PDE: the **McKean-Vlasov equation**



Photo © Dave Conner (CC BY 4.0) – clip-art adaptation by L. Moschen.

The McKean-Vlasov Equation

$$\partial_t \mu = \underbrace{\sigma \Delta \mu}_{\text{Diffusion}} + \underbrace{\nabla \cdot (\mu \nabla V)}_{\text{External drift}} + \underbrace{\nabla \cdot (\mu \nabla W * \mu)}_{\text{Interaction drift}}^{12}$$

- $\mu(t, x)$: evolution of the density of agents
- $W(x)$: introduces **nonlinearity** and **nonlocality** through $(\nabla W * y)(x) = \int_{\Omega} \nabla W(x - t) y(t) dt$
- Fokker-Planck PDE for the McKean-Vlasov SDE where the drift depends on the law μ

¹A.-S. Sznitman, Topics in propagation of chaos, Lecture Notes in Mathematics, vol. 1464, Springer, 1991.

²J.A. Carrillo, R.J. McCann, and C. Villani, Kinetic equilibration rates for granular media and related equations, Rev. Mat. Iberoamericana, 2003.

Why Control Interacting Systems?

Motivations for feedback design

- Uncontrolled systems:
 - Possible **slow convergence** to the long-term (small spectral gap)
 - Converge to an **undesirable equilibrium**
- Our goals:
 - **Accelerate convergence** to a desired steady state
 - **Steer the system** toward or away from specific modes
- Solution approach:
 - Introduce **time-dependent feedback potentials** into the PDE dynamics
 - Based on Breiten, Kunisch, and Pfeiffer's work³

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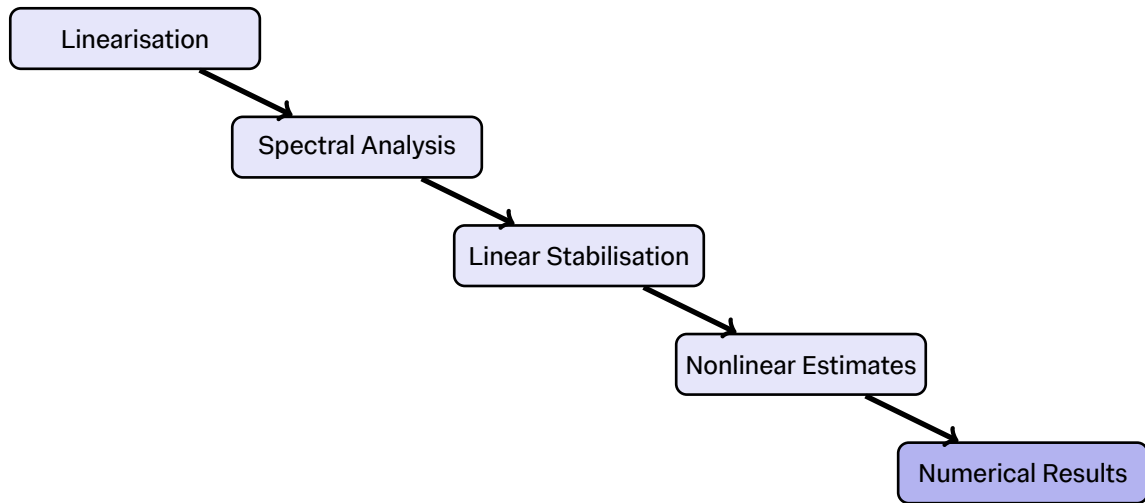
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Overview of Our Approach



From Interacting Particles to the Mean-Field PDE

- **Agent-based model:** each of N agents evolve according to

$$dX_i(t) = -\nabla V(X_i) dt - \frac{1}{N} \sum_{j=1}^N \nabla W(X_i - X_j) dt + \sqrt{2\sigma} dB_i(t)$$

- **Mean-field limit** $N \rightarrow \infty$: overdamped regime

$$\partial_t \mu = \nabla \cdot (\sigma \nabla \mu + \mu (\nabla V + \nabla W * \mu)), \quad t > 0, x \in \Omega$$

- **Conditions:**

- We consider $\Omega = \mathbb{R}^n$, with decay, or $\Omega = \mathbb{T}^n$, with periodic boundary conditions
- We ask $W \in W^{2,\infty}(\Omega)$ and $V \in C^2(\Omega)$. If $\Omega = \mathbb{R}^n$,

$$\lim_{|x| \rightarrow \infty} \frac{1}{4\sigma} |\nabla V(x)|^2 - \frac{1}{2} \Delta V(x) = \infty$$

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Long-Time Behaviour of the McKean-Vlasov Equation

- **Steady states:** time-independent solutions $\bar{\mu}$ satisfying

$$\nabla \cdot (\sigma \nabla \bar{\mu} + \bar{\mu}(\nabla V + \nabla W * \bar{\mu})) = 0 \implies \bar{\mu} \propto \exp \left\{ -\frac{1}{\sigma} (V + W * \bar{\mu}) \right\}$$

- **Gradient flow:** the dynamics decrease a free energy functional. Steady states are critical points of this energy
- **Convex potentials:** convexity of V and W implies unicity of $\bar{\mu}$ and $\mu(t, \cdot) \xrightarrow{L^1} \bar{\mu}$ exponentially⁴
- **Nonconvex potentials:** multiple steady states may emerge.

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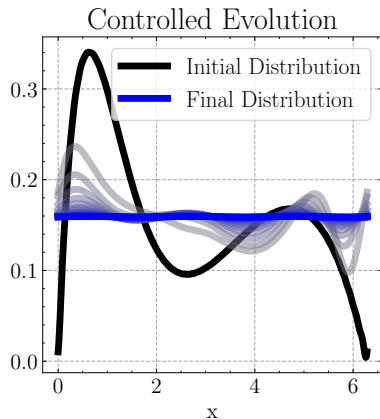
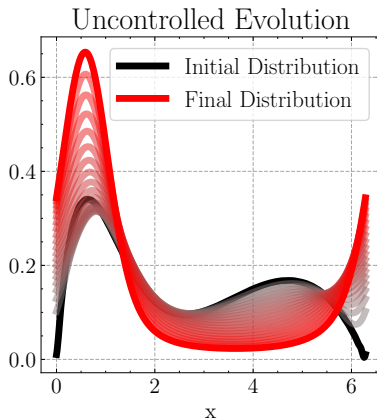
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Noisy Kuramoto Model

What kind of problem?

- Coupled phase oscillators with sine interaction that synchronize⁵
- Let $\Omega = [0, 2\pi)$, $V(x) = 0$, $W(x) = -K \cos(x)$
- If $K \leq 1$: $\bar{\mu}(x) = \frac{1}{2\pi}$ is the unique steady state
- If $K > 1$: $\bar{\mu} = \frac{1}{2\pi}$ becomes unstable and infinite new steady states appear



⁵L. Bertini, G. Giacomini, K. Pakdaman. Dynamical Aspects of Mean Field Plane Rotators and the Kuramoto Model, J Stat Phys. 2010.

Formulation of the Controlled Dynamics

- **Objective:** accelerate convergence to $\bar{\mu}$ or stabilise otherwise unstable equilibria
- **Control strategy:** modify the potential as

$$V(\mathbf{x}) \mapsto V(\mathbf{x}) + \overbrace{\sum_{j=1}^m \mathbf{u}_j(\mathbf{t}) \alpha_j(\mathbf{x})}^{\text{Control input}},$$

for **chosen** spatial controls α_j and **to be optimized** time-dependent controls $\mathbf{u}_j$

- **Controlled PDE:**

$$\partial_t \mu = \nabla \cdot \left(\mu \left(\nabla V + \nabla W * \mu + \sum_{j=1}^m \mathbf{u}_j \nabla \alpha_j \right) \right) + \sigma \Delta \mu$$

⁶T. Breiten, K. Kunisch, and L. Pfeiffer. Control strategies for the Fokker-Planck equation. ESAIM: COCV, 2018.

Operator-Theoretic Reformulation

- **Rewriting the equation:** let $y = \mu - \bar{\mu}$, then

$$\dot{y} = \overbrace{(\mathcal{A} + \mathcal{D}_W)y}^{\text{linear part}} + \sum_{j=1}^m \overbrace{u_j(t) \mathcal{N}_j y}^{\text{bilinear control}} + \sum_{j=1}^m \overbrace{\mathcal{B}_j u_j(t)}^{\text{pure input term}} + \overbrace{\mathcal{W}(y)}^{\text{nonlinear remainder}}, \quad \mathcal{B}_j = \mathcal{N}_j \bar{\mu}$$

- **Operators:** $\mathcal{A}\mu = \nabla \cdot (\sigma \nabla \mu + \mu \nabla V)$, $\mathcal{N}_j \mu = \nabla \cdot (\mu \nabla \alpha_j)$, $\mathcal{W}(\mu) = \nabla \cdot (\mu \nabla W * \mu)$

- **State space:** $\mathcal{X} = L^2(\Omega, \bar{\mu}^{-1}) := \left\{ f : \int_{\Omega} |f(x)|^2 \bar{\mu}^{-1}(x) dx < \infty \right\}$

- **Linearization:** the Fréchet derivative $\mathcal{D}_W$ of $\mathcal{W}$ at $\bar{\mu}$ is

$$\mathcal{W}(\bar{\mu} + y) = \underbrace{\nabla \cdot \left[\overbrace{\bar{\mu}(\nabla W * y)}^{\text{nonlocal part } \mathcal{D}_{W,2}} + \overbrace{y(\nabla W * \bar{\mu})}^{\text{local part } \mathcal{D}_{W,1}} \right]}_{\mathcal{D}_W y} + \underbrace{\nabla \cdot [y(\nabla W * y)]}_{\mathcal{W}(y)}$$

How do we plan to control?

- **Goal:** Stabilise an equilibrium $\bar{\mu}$ of the McKean–Vlasov equation via **feedback control**.
- **Linearised dynamics around $\bar{\mu}$:**

$$\dot{y} = (\mathcal{A} + \mathcal{D}_W)y + \sum_{j=1}^m \cancel{u_j(t) \mathcal{N}_j} y + \sum_{j=1}^m \mathcal{B}_j u_j(t) + \cancel{\mathcal{W}(y)},$$

where $\mathcal{B}_j = \nabla \cdot (\bar{\mu} \nabla \alpha_j)$. Fix $\delta > 0$. We **have to** choose α_j .

- **Key idea:** control the **directions** related to eigenvalues with real part $\geq -\delta$.
- **Strategy:**



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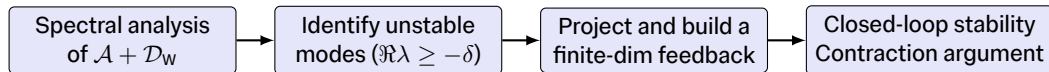
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Spectral Reformulation via Unitary Transformation

- **Ground-state transform:** the unitary map

$$\mathcal{U} : L^2(\Omega, \bar{\mu}^{-1}) \rightarrow L^2(\Omega), \quad \mathcal{U}y = \frac{y}{\sqrt{\bar{\mu}}}$$

- **Transformed operator:** the linearised operator becomes

$$\mathcal{H} := -\mathcal{U}(\mathcal{A} + \mathcal{D}_W)\mathcal{U}^{-1} = \underbrace{-\sigma\Delta + \underbrace{\Psi(x)}_{\Psi = \frac{1}{4\sigma}|\nabla V|^2 - \frac{1}{2}\Delta V}}_{\text{Schrödinger operator}} + \underbrace{\mathcal{K}}_{\text{Hilbert-Schmidt operator}}$$

- **Spectral structure:**

- Rich spectral theory and numerical methods for Schrödinger operators
- $\mathcal{H}$ has compact resolvent and a pure point spectrum⁷
- $\sigma(\mathcal{H}) = -\sigma(\mathcal{A} + \mathcal{D}_W)$, with isolated eigenvalues **accumulating at infinity**
- There are **finitely many eigenvalues** with $\Re(\lambda) \geq -\delta \rightarrow$ we control these modes

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⁷M. Reed and B. Simon. Analysis of Operators; Methods of Modern Mathematical Physics IV. 1978.

Mass Conservation and Zero-Mean Projection

- **Observation:** the operator $\mathcal{A} + \mathcal{D}_W$ preserves mass:

$$\int_{\Omega} (\mathcal{A} + \mathcal{D}_W)y(x) \, dx = \int_{\Omega} \nabla \cdot (\sigma \nabla y(x) + y(x) \nabla (V + W * \bar{\mu})(x) + \bar{\mu}(x) (\nabla W * y)(x)) \, dx = 0,$$

so $0 = \langle (\mathcal{A} + \mathcal{D}_W)y, \bar{\mu} \rangle_{L^2(\Omega, \bar{\mu}^{-1})}$ for all $y \in L^2(\Omega, \bar{\mu}^{-1})$

- **Consequence:** $(\mathcal{A} + \mathcal{D}_W)^* \bar{\mu} = 0 \implies 0$ is an eigenvalue of $\mathcal{A} + \mathcal{D}_W$ we don't need to control
- **Fix:** we work in the subspace of zero-mean perturbations:

$$\mathcal{X}_0 := \left\{ y \in L^2(\Omega, \bar{\mu}^{-1}) : \int_{\Omega} y(x) \, dx = 0 \right\}$$

- **Notation:** for presentation purposes, we keep writing $\mathcal{A}$ and $\mathcal{D}_W$ but implicitly restrict them to $\mathcal{X}_0$

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Optimal Control and Stabilisation of the Linearised Problem

Consider the problem

$$\min_{\mathbf{u}(\cdot)} \frac{1}{2} \int_0^\infty e^{2\delta t} \left(\langle \mathbf{y}(t), \mathcal{M} \mathbf{y}(t) \rangle + \|\mathbf{u}(t)\|^2 \right) dt, \quad \mathcal{M} \text{ positive definite}$$

subject to $\dot{\mathbf{y}}(t) = \underbrace{(\mathcal{A} + \mathcal{D}_w)}_{=: \mathcal{L}} \mathbf{y}(t) + \underbrace{[\mathcal{B}_1, \dots, \mathcal{B}_m]}_{=: \mathcal{B}} \mathbf{u}(t),$

- **Riccati equation:** find a self-adjoint operator Π solving

$$(\mathcal{L}^* + \delta I)\Pi + \Pi(\mathcal{L} + \delta I) - \Pi \mathcal{B} \mathcal{B}^* \Pi + \mathcal{M} = 0$$

- **Feedback law:** $\mathbf{u}(t) = -\mathcal{B}^* \Pi \mathbf{y}(t)$ stabilises the system, i.e., $\|\mathbf{y}\|_{L^2(\Omega, \bar{\mu}^{-1})} \leq C e^{-\delta t}$. We need to verify the **infinite-dimensional Hautus condition**⁸

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⁸R. Curtain and H. Zwart, An Introduction to Infinite-Dimensional Linear Systems Theory, Springer-Verlag, 2005.

How to choose $\mathcal{B}_j$ appropriately?

- **Eigen-expansion:**

$$y(t, \mathbf{x}) = \sum_{n \geq 1} c_n(t) \varphi_n(\mathbf{x}), \quad (\mathcal{A} + \mathcal{D}_W)\varphi_n = \lambda_n \varphi_n, \quad \langle \varphi_n^*, \varphi_m \rangle = \delta_{nm}$$

- **Mode dynamics:**

$$\dot{c}_n = \lambda_n c_n + \sum_{j=1}^m u_j(t) \langle \mathcal{B}_j, \varphi_n^* \rangle$$

- **Design inputs to target unstable modes:** Solve

$$\mathcal{B}_j := \nabla \cdot (\bar{\mu} \nabla \alpha_j) = \varphi_j, \quad j = 1, \dots, m$$

- **Decoupled slow block:** For the m modes with $\Re(\lambda_j) \geq -\delta$, we get

$$\dot{c}_j = \lambda_j c_j + u_j, \quad j = 1, \dots, m$$

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$$\dot{c}_n = \lambda_n c_n + \sum_{j=1}^m u_j(t) \langle \mathcal{B}_j, \varphi_n^* \rangle$$

- **Design inputs to target unstable modes:** Solve

$$\mathcal{B}_j := \nabla \cdot (\bar{\mu} \nabla \alpha_j) = \varphi_j, \quad j = 1, \dots, m$$

- **Decoupled slow block:** For the m modes with $\Re(\lambda_j) \geq -\delta$, we get

$$\dot{c}_j = \lambda_j c_j + u_j, \quad j = 1, \dots, m$$

How to choose $\mathcal{B}_j$ appropriately?

- **Eigen-expansion:**

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$\mathcal{B}_j = \varphi_j \implies$ infinite-dimensional Hautus condition $\implies \delta$ -stability of the linearised system

Local Exponential Stabilisation: Closed-Loop View

The equation for $\psi = e^{\delta t} \mathcal{U}y$ is

$$\dot{\psi} = -\mathcal{H}_{\Pi}\psi - (\hat{\mathcal{B}}^* \hat{\Pi} \psi) \hat{\mathcal{N}}^{\delta}(\mathbf{t})\psi + \hat{\mathcal{W}}^{\delta}(\psi), \quad \psi(0, \cdot) = \mathcal{U}(\mu_0 - \bar{\mu})$$

where $\mathcal{H}_{\Pi} = \mathcal{H} - \delta \mathbf{I} + \hat{\mathcal{B}} \hat{\mathcal{B}}^* \hat{\Pi}$.

- **Linear decay:** $\mathcal{H}_{\Pi}$ generates exponential decay with rate δ
- **Nonlinear remainder:** the term $\hat{\mathcal{W}}^{\delta}(\psi)$ allows a Lipschitz-type estimate
- **Contraction argument:** for $\|\psi_0\|$ sufficiently small, the full closed-loop system is a **contraction** in a suitable function space⁹. Therefore,

$$\|\psi(\mathbf{t})\|_{L^2(\Omega)} \leq \mathbf{C} \|\psi_0\|_{L^2(\Omega)} \implies \|y(\mathbf{t})\|_{L^2(\Omega, \bar{\mu}^{-1})} \leq \mathbf{C} e^{-\delta \mathbf{t}} \|\mu_0 - \bar{\mu}\|_{L^2(\Omega, \bar{\mu}^{-1})}.$$

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⁹D. Kalise, L. M, G. Pavliotis. Linearization-Based Feedback Stabilization of McKean-Vlasov PDEs, arXiv. 2025.

Numerical implementation

Work on the periodic domain $\Omega = \mathbb{T}^d$

- **Fourier expansion:** for $k \in \mathbb{Z}^d$,

$$y(x, t) \approx \frac{1}{\sqrt{2\pi}} \sum_{|k| \leq L} \hat{y}_k(t) e^{ik \cdot x}$$

- **Galerkin projection:** project the full equation onto the span of $\left\{ \frac{1}{\sqrt{2\pi}} e^{ik \cdot x} \right\}_{|k| \leq L}$. The matrix representation is

$$\dot{\hat{y}} = A\hat{y} + \sum_{j=1}^m B_j u_j, \quad J = \int_0^\infty e^{2\delta t} (\hat{y}^\top M \hat{y} + \|u\|^2) dt$$

- **Feedback computation:** solve the discrete Riccati equation

$$(A^\top + \delta I)\Pi_L + \Pi_L(A + \delta I) - \Pi_L B B^\top \Pi_L + M = 0, \quad u = -B^\top \Pi_L \hat{y}$$

Noisy Kuramoto under feedback control

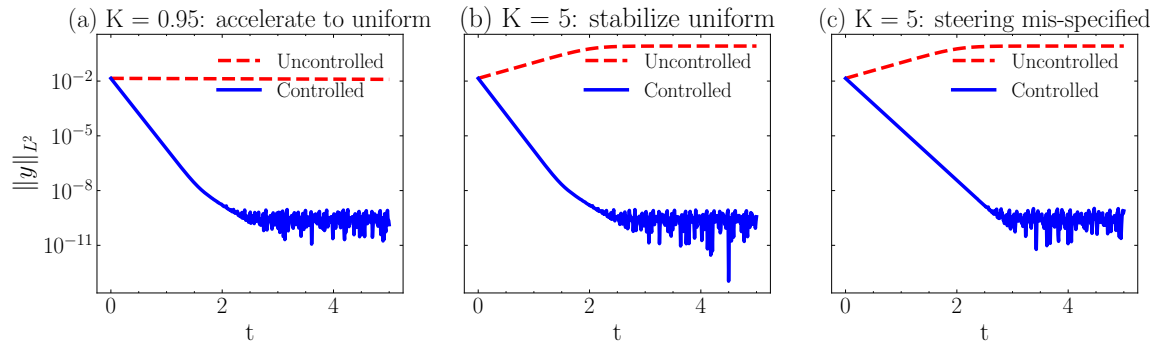


Figure: Set $V(x) = 0$, $W(x) = -K \cos(x)$ and $\sigma = 0.5$.

Kuramoto model + Potential V

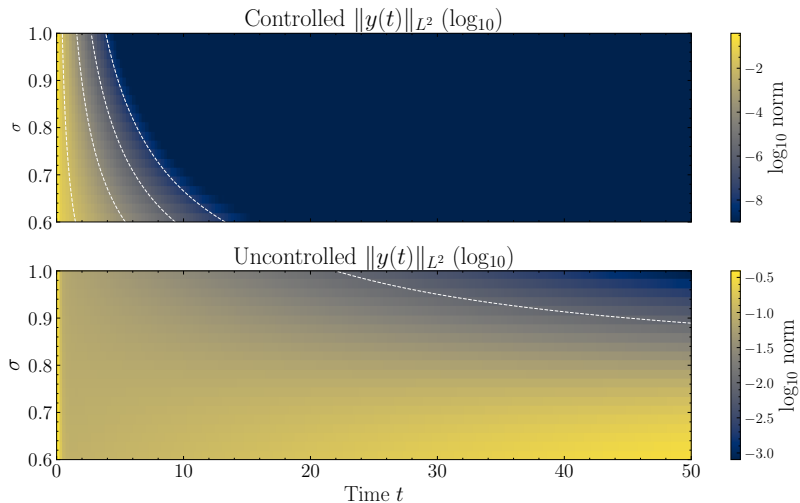


Figure: Set $V(x) = 0.05 \cos(x)$ and $W(x) = -\cos(x)$

Outlook

Conclusions

- Deterministic feedback accelerates convergence to steady states.
- Riccati-based law yields local, rate-guaranteed stabilization.

Future Work

- Develop a global Lyapunov feedback law for full nonlinear PDE.
- Analyze robustness under model uncertainties in V and W .
- Numerics to higher-dimensional domains.
- Extend to kinetic equations: hypoelliptic analysis

IMPERIAL

Thank you for the attention!
Questions, suggestions?

Feedback stabilisation for the McKean-Vlasov equation
July 17th, 2025

Operator-Theoretic Reformulation

Weighted spaces and linearisation

- **Why work in weighted spaces?**

- This is the natural energy space for the linearised and localised operator

$$(\mathcal{A} + \mathcal{D}_{W,1})y := \nabla \cdot (\sigma \nabla y + y \nabla V) + \nabla \cdot (y \nabla W * \bar{\mu})$$

- It ensures that the operator $\mathcal{A} + \mathcal{D}_{W,1}$ is self-adjoint.
- Inner product structure simplifies spectral and stability analysis.
- By a unitary transformation, we convert $\mathcal{A} + \mathcal{D}_W$ into a Schrödinger operator plus a compact operator.

- **Two linearisation strategies:**

One can adopt the full linearisation $\mathcal{D}_W$, which captures all nonlocal effects, or the simplified local form $\mathcal{D}_{W,1}$, which is easier to handle computationally. Both yield implementable schemes.

$$\overbrace{\sigma \Delta y + \nabla \cdot (y \nabla (V + W * \bar{\mu}))}^{\text{Linear Fokker-Planck operator}} + \nabla \cdot (\bar{\mu} \nabla W * y)$$